

Paving the Way for Next-Generation All-Solid-State Batteries: Dry Electrode Technology

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For the effective implementation of all-solid-state batteries (ASSBs), the progress of dry electrode technology is essential. Considering the urgent challenges posed by global warming, advancing affordable ASSBs is crucial for reliable and sustainable electrochemical energy conversion and storage systems. Despite the substantial progress in research on solid electrolytes, ASSBs continue to face fundamental challenges, including high manufacturing costs, considerable environmental impact, and insufficient interfacial stability. The dry electrode processing methodology presents a highly promising alternative to conventional wet electrode fabrication techniques, effectively addressing these limitations while facilitating scalable production capabilities. Although research on dry-electrode-based ASSBs has gained significant traction in recent years, the overall challenges are recognized as intricate due to the presence of multiple electrode components with diverse requirements and limitations. For the comprehensive review, the dry processing approach is systematically examined from the perspective of materials science and electrode processing. An electrode-component-level framework is adopted as well. Furthermore, crucial technical barriers are identified, and prospective research directions for the advancement of solid-state battery technology are delineated.

warming and the increasing electricity demand by the exponential growth of artificial intelligence and machine learning. The rapid expansion and huge applications of LIB have prompted a paradigm shift in the performance requirement, expanding conventional metrics to include safety, environmental sustainability, and economic feasibility. However, LIB components, such as active materials and flammable liquid electrolytes, have remained largely unchanged since the inception of LIBs.^[1] As large-scale LIBs have emerged as conventional technology in applications such as electric vehicles (EVs) and energy storage systems (ESSs), safety concerns have been recognized as a critical challenge. In fact, safety considerations in LIBs have received comparatively less due to the exceptionally low defect rates achieved through advanced manufacturing processes for current-level LIBs. However, the ongoing pursuit of high energy density within constrained volume has inevitably diminished the safety margin. The rising financial burden associated with battery-related incidents has

further elevated safety as a critical concern. Consequently, safety has re-emerged as a paramount concern, necessitating the development of more stable LIB configurations at the fundamental component level.

1. Introduction

Lithium-ion batteries (LIB) have become integral for energy conversion and storage in response to the accelerating global

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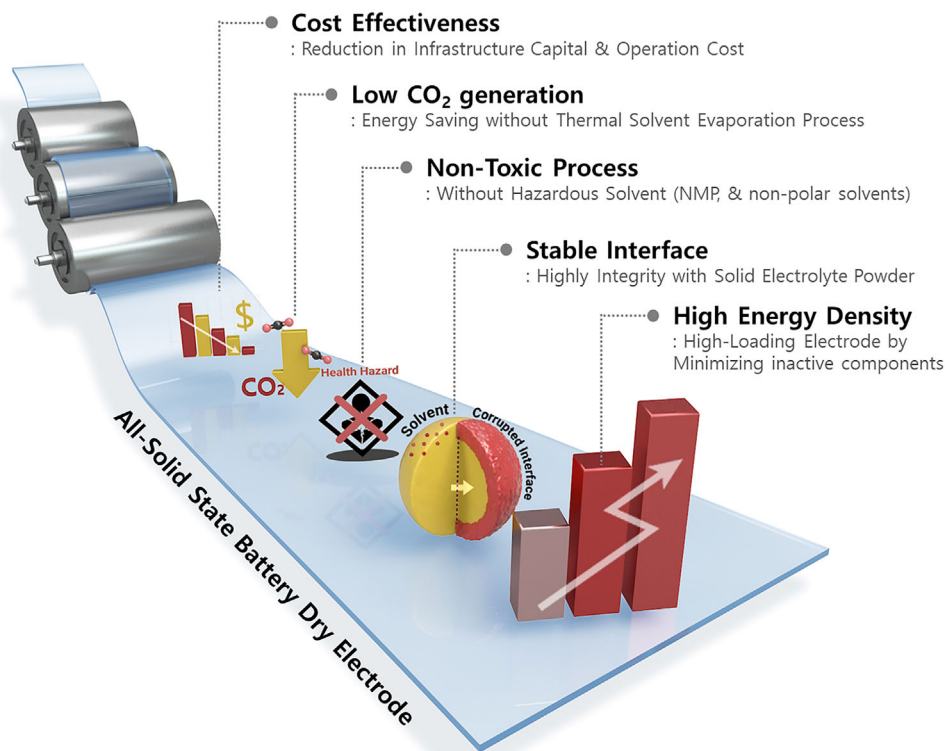


Figure 1. Schematic illustration highlighting the main benefits of the dry electrode process for ASSBs.

All-solid-state batteries (ASSBs) offer a promising solution to these safety challenges by employing solid-state electrolytes that eliminate the vapor pressure and flammability risks associated with liquid organic electrolytes. Recent breakthroughs in high-conductivity solid electrolytes, particularly sulfide-based materials such as $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSCl) and $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS), have enhanced the commercial viability of ASSBs, leading to a surge in this area.^[2–5] Such materials achieve ionic conductivities comparable to those of liquid electrolytes, making ASSBs a next-generation technology that integrates enhanced safety and high energy density. ASSBs enable the incorporation of high-energy-density active materials, such as lithium (Li) metal, while minimizing the reliance on supporting components related with safety used in conventional LIBs, modules, and packs.^[6,7] These two significant advantages of ASSBs have accelerated their utilization, enabling breakthroughs in electrochemical energy conversion and storage systems.

However, ASSBs face several manufacturing issues by complex constraints including improper electrode microstructure, high production costs, and environmental burdens. The unique properties of solid electrolytes necessitate novel approaches to electrode manufacturing. Although sulfide-based solid electrolytes offer superior ionic conductivity, they also present hurdles, including interfacial stability, particle mixing, and crystallinity effects, all of which significantly influence battery performance. Additionally, sulfide electrolytes are highly reactive with conventional electrode slurry solvents, raising concerns regarding interfacial degradation and the environmental impact of using toxic solvents in the fabrication process. The primary ki-

netic limitations arise from improperly engineered electrode microstructure and the high interfacial resistive behavior among particle contacts.^[8–10] For high electrochemical performance, both ionic and electronic pathways must be efficiently established through a homogeneous distribution of active material, conducting agent, binder, and solid electrolyte in the electrode. Conventional electrodes for LIBs have been fabricated by wet-electrode methods via slurry preparations and drying processes. Due to the uncertainty associated with rheological constraints during slurry mixing and the complex multi-phase behaviors during the drying process, achieving an ideal electrode microstructure is a challenge. Interfacial failure, conducting agent aggregations, and intricate ionic diffusion pathways further complicate microstructural control.^[11–15] The electrode fabrication process for ASSBs is a critical challenge, as the absence of liquid electrolytes leads to fundamentally different electrochemical microstructure compared to systems with facile ionic diffusion via liquid electrolytes. These manufacturing hurdles limit the use of ASSBs and prevent them from achieving the performance levels of state-of-the-art LIBs.

To address these challenges, the dry electrode process is a promising solution for ASSBs (Figure 1). A key advantage of this approach is its cost-effectiveness because it eliminates the energy-intensive drying step required for conventional wet-electrode processing. The infrastructure handling organic solvent is not necessary such as the drying section of the coating line, solvent recovery systems, and circulating management, thereby reducing both capital and operational expense. Notably, during battery manufacturing, the drying step with a

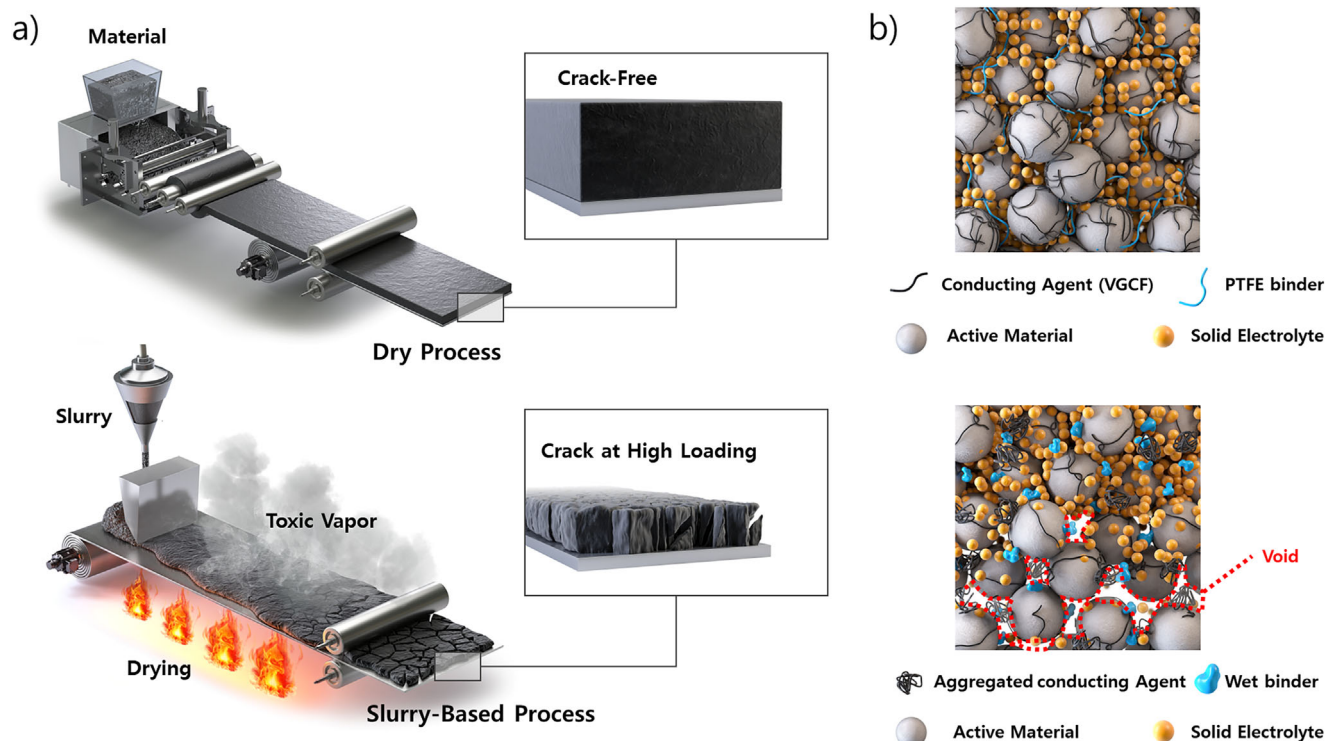


Figure 2. a) Comparison of the dry (upper) and wet coating (lower) electrode processes. b) Microscopic conceptual images of the cross-sectioned dry (upper) and wet-coated (lower) electrodes.

high-temperature sequence consumes the highest amount of energy, and its elimination can significantly reduce the overall energy footprint and associated CO₂ emissions.^[1,16] The solvent-free nature of the dry electrode process enhances environmental sustainability by reducing energy consumption through the removal of the drying step. For routine slurry fabrication with sulfide solid electrolytes, limited types of solvents that are nonpolar or less polar, such as toluene, xylene, and isopropyl butylate, are used. However, they are usually more toxic and expensive than conventional organic solvents, such as *N*-methylpyrrolidinone (NMP) used for the LIB manufacturing process.^[17–21] The dry electrode process circumvents these issues while simultaneously improving the interfacial stability of solid electrolytes. The ideal electrode microstructure integrates the totally homogeneous mixture among active material, binder, and conducting agent. However, the manufactured electrodes have defect microstructures such as aggregation, and collapse of electron pathway increasing polarization.^[7,13] The reactive nature of sulfide electrolytes causes a surface passivation layer, accelerating a high interfacial resistance in contact with the slurry solvent in the wet-processing methods. This challenge is mitigated using the dry-coating process by waiving the step of contacting fragile solvents in electrode manufacturing. Additionally, the dry electrode process is well-suited for high-loading electrode production, facilitating the fabrication of thicker electrodes without the constraints associated with wet processing. A thick electrode structure is effective in minimizing the ratio of inactive supporting components such as the current collector and electrolyte sheet in ASSBs. As the thickness of the electrode increases, the com-

plexity of drying from the wet-processed thick electrodes needs a longer drying step. It harms the surface integrity of solid electrolyte limiting ionic/electronic conducting microstructure. Finally, the dry-electrode process is beneficial to fabricate credible thick electrodes for increasing the energy density of ASSBs.

A comparative summary of electrode processing methods, including their respective advantages and limitations, is presented in **Figure 2**. Dry-processed electrodes offer superior formability for maintaining the designed electrode structure and can be fabricated through a facile combination of mixing and calendaring processes (Figure 2a). Conversely, wet-processed electrodes require an additional drying step because the electrode slurry must be dried to form the final electrode structure. This additional processing step increases the overall complexity and burden of the electrode fabrication process. For instance, increasing the temperature during the drying process can shorten the drying time, however, increase the reactivity of solid electrolytes. In this regard, the overall drying duration of the wet process in ASSBs may increase. If the electrode remains in a fluidic state during processing, the slurry may flow, making it challenging to maintain the intended electrode dimensions. This issue becomes more critical in the fabrication of thick electrodes, further prolonging the drying process. Additionally, the drying step consumes additional thermal energy, with the evaporation of the nonreactive solvent in the slurry raising environmental concerns. At the microscopic level, in wet-processed electrodes, sudden or inhomogeneous solvent evaporation during drying can lead to pore formation within the electrode by repulsing particles (Figure 2b). Moreover, the uppermost region of the electrode, which

experiences the highest tensile stress, is particularly susceptible to crack formation as drying and particle distribution progress. The dry-processed electrodes retain their physically homogeneous mixture, ensuring uniform dispersion of multicomponent systems with active materials, conductive agents, binders, and solid electrolytes throughout the electrode. The electrode fabrication process is highly complicated in terms of dry granular flow interacting among electrode components. As the selection and combination of various electrode components varies in purpose of ASSB performance, the systematic approaches by electrode components are essential to deliver credible dry electrode processes.

The integration of the dry electrode process with the ASSB technology marks a pivotal advancement in the development of solid-state batteries, improving manufacturing feasibility while reducing costs and increasing processing flexibility. This technological convergence promises to unlock the full potential of high-energy-density, environmentally sustainable batteries, potentially transcending the limitations of conventional rechargeable batteries and significantly contributing to global energy security and sustainability. In this review, we comprehensively examine the academic and technological challenges associated with dry electrode fabrication for ASSBs and systematically explore various aspects of the dry electrode technology, including material science, fabrication processes, electrode structural characteristics, electrochemical performance, and commercialization challenges. Particular attention is focused on the complex microstructures among active materials, solid electrolyte powder, conducting agents, and binders, as these internal behaviors predominantly influence the electrochemical performance of ASSBs. A component-based review is essential to unveil the synergistic or detrimental effects arising from these microstructures under the dry electrode process.^[11–15]

2. Dry Coating Process for ASSBs

To achieve high-energy-density ASSBs with superior electrochemical performance, it is crucial to enhance the Li-ion (Li^+) kinetics and electrochemical stability of both the electrode and solid electrolyte membrane.^[22] From the perspective of the material, among the different components of ASSBs, such as the cathode active material, binder, conducting agent, and solid electrolyte, the intrinsic properties of the solid electrolyte play a pivotal role in determining the Li^+ conductivity and electrochemical stability.^[23,24] Various approaches, including engineering of the chemical composition (e.g., stoichiometric ratios, doping, and surface coating) and/or particle morphology (e.g., particle size and distribution) for the solid electrolyte, have been explored to enhance the Li^+ conductivity and electrochemical stability in the cell. However, beyond the intrinsic properties of the materials, the interfacial characteristics between the electrode components or the electrode and solid electrolyte membrane also play an important role in the kinetics of the electrons and/or Li^+ .^[25]

In the ASSB electrode, a greater extent of solid electrolyte coverage over the active material surface enhances the Li^+ conductivity owing to the enhanced Li^+ flux at the interface. Moreover, the distribution of other components, such as conducting agents and binders within the cathode, affects not only the electron conductivity but also the Li^+ conductivity.^[26] The formation of an in-

timite interface between the cathode layer and solid electrolyte membrane is also crucial for suppressing the degradation of the Li^+ conductivity and the corresponding electrochemical properties. The poor adhesion between these layers can result in the formation of voids at the interface.^[27] Therefore, a homogeneous distribution of the components within the cathode or solid electrolyte membrane through engineering the microstructure of these layers should be achieved to improve the Li^+ conductivity and, consequently, the overall electrochemical performance of ASSBs. The microstructures of both the electrode and solid electrolyte membrane can be engineered during the manufacturing process.

The following sections explore the ways the manufacturing processes of the electrode and solid electrolyte membrane, including the wet and dry coating processes, affect their physicochemical properties, such as their microstructure, chemical stability, and mechanical and electrochemical performance. Subsequently, the relationship between the properties of the electrode and solid electrolyte is discussed. Finally, strategies to enhance the energy density of ASSBs are examined, focusing on reducing the amounts of inactive materials while increasing the number of active layers, which is directly related to cell design and configuration.

2.1. Electrode Microstructures Prepared by the Wet and Dry Coating Processes

The wet coating process for the preparation of both electrodes and solid electrolyte membranes results in binder migration due to capillary forces during solvent drying in the LIB and ASSB systems.^[28] Binder migration induces compositional gradients and porosity differences between the top and bottom regions of the electrode or solid electrolyte membrane. This inhomogeneous microstructure degrades the electrochemical behavior. Furthermore, the deficient binder content in the bottom region weakens the adhesion strength between the electrode and current collector, potentially causing delamination, which worsens with increasing electrode thickness.^[29] The highly concentrated binder content in the top region can induce significant degradation in the kinetics of both the electrons and Li^+ . Electrodes prepared via the wet and dry coating processes exhibit significantly different microstructures, especially in the distribution of the electrode components (**Figure 3a**). During the wet coating process, binders spread widely on the surface of the cathode active materials, and conducting agents tend to aggregate, inducing voids around the surface of the cathode active materials. These voids facilitate the loss of contact between the solid electrolyte and active materials, lowering the coverage of the solid electrolyte over the cathode active materials, and thereby lowering the Li^+ diffusion and conductivity between the solid electrolyte and active materials. Conversely, in the dry electrode-coating process, the binder coating on the surface of the cathode active materials is limited.^[30] During the dry mixing process in the dry coating process, a shear force is applied to the electrode components, which induces deformation of the solid electrolyte, increasing its coverage over the cathode active material, and thereby improving the Li^+ diffusion pathway and conductivity.^[31] Moreover, wet coating processes often require a higher binder content to ensure an

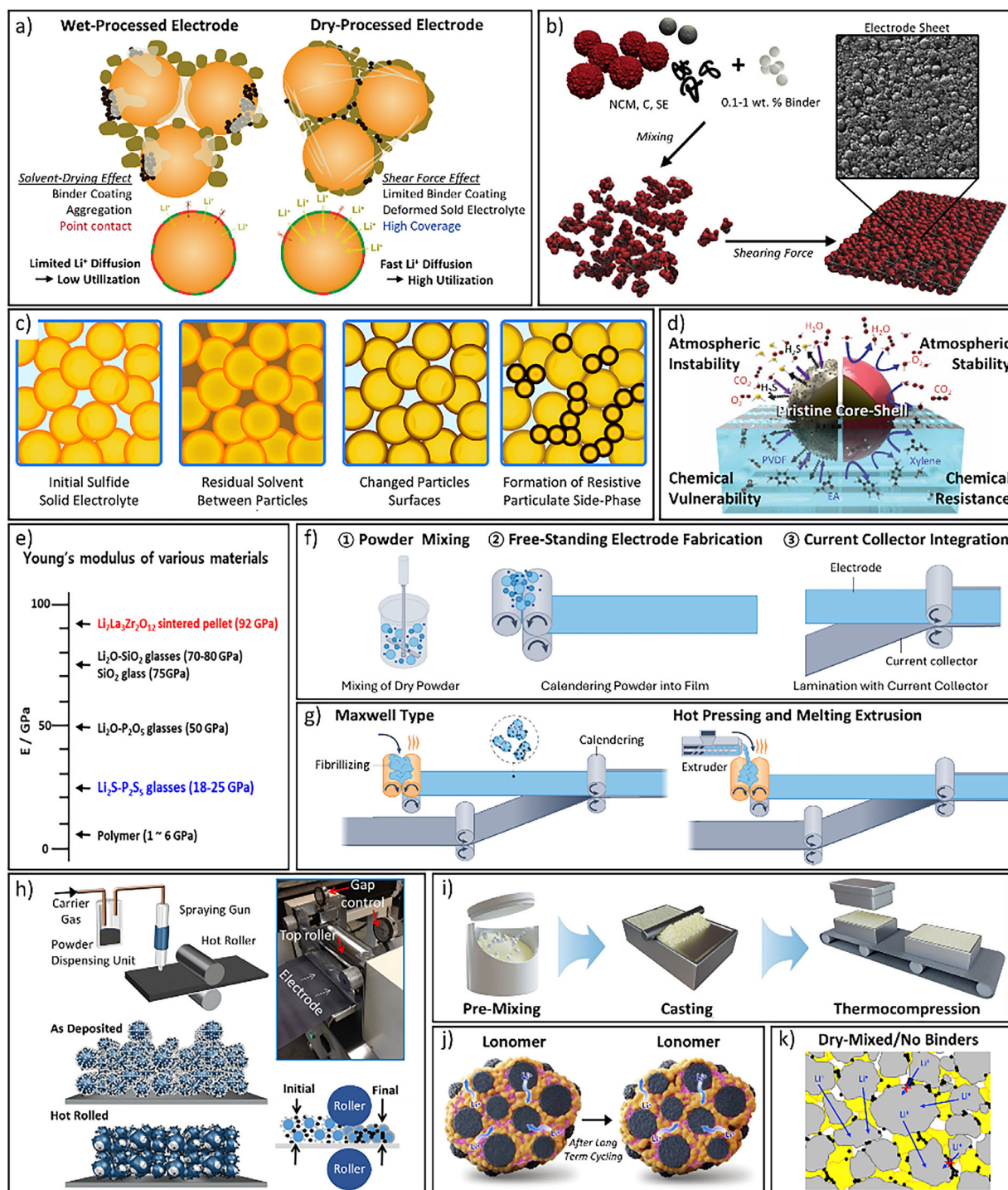


Figure 3. Electrode microstructures are prepared via the wet and dry coating processes together with the binders and methods used in the dry electrode coating process. a) Microstructural differences between the wet and dry coating processes. Reproduced with permission.^[31] Copyright 2024, American Chemical Society. b) Preparation of a dry-processed cathode with a low PTFE binder content (0.1–1 wt.%). Reproduced with permission.^[33] Copyright 2019, Elsevier B.V. All rights reserved. c) Chemical stability of a solid electrolyte with solvent during the wet coating process. Reproduced with permission.^[36] Copyright 2022, Royal Society of Chemistry. d) Strategies for the suppression of side reactions between solid electrolytes and solvents. Reproduced with permission.^[38] Copyright 2020, American Chemical Society. e) Young's modulus for various materials.^[40] f) The main steps of the dry coating process and g) two representative types of the roll-to-roll dry coating process: the Maxwell type method (left) and the extrusion type method (right). Reproduced with permission.^[49] Copyright 2025, Springer Nature. h) Electrostatic spraying method using a PVDF binder for the dry coating process. Reproduced with permission.^[53] Copyright 2016, Springer Nature. i) Fusion bonding technique for the dry coating process to fabricate the composite sulfide film. Reproduced with permission.^[52] Copyright 2024, Wiley-VCH. j) Alternative binders and k) binder-free systems for the dry coating process. Reproduced with permission.^[55] Copyright 2022, American Chemical Society and ^[56] Copyright 2017, Elsevier B.V.

adequate binding of the electrode components. However, polymeric binders inherently act as insulators, reducing both the Li^+ and electron conductivities within the electrode. In contrast, the dry electrode-coating process typically employs polytetrafluoroethylene (PTFE) as a binder, which is stretched into fibrous structures under shear force. The deformation and fibrillization of the PTFE binders via a shear force afford a fibrous network structure within the electrode, enabling superior binding properties even with a low binder content.^[32] Hippauf et al. demonstrated that dry-processed cathodes with a low binder content (0.1–1 wt.%) could deliver both superior binding characteristics and electrochemical performance (Figure 3b).^[33] These findings underscore the significance of the dry electrode process in manufacturing electrodes or solid electrolyte membranes with high mechanical strength and superior interfacial properties associated with a homogeneous microstructure of the layers.

The occurrence of undesirable side reactions between the solid electrolytes and solvents is another critical issue in the wet coating process.^[34] The solid electrolytes used in ASSBs are typically categorized into oxide-, sulfide-, and halide-based electrolytes. Oxide solid electrolytes suffer from poor ductility and hinder contact with the active materials of the cathode, while halide electrolytes exhibit low stability and high reactivity with Li .^[35] Consequently, sulfide-based electrolytes have garnered considerable attention owing to their balanced properties. However, sulfide-based electrolytes can interact with solvents during the wet coating process, producing byproducts such as H_2S , PO_4^{3-} , SO_4^{2-} , and LiOH , which lead to severe structural degradation (Figure 3c).^[36] The formation of these byproducts increases the interfacial resistance and diminishes the Li^+ conductivity, ultimately degrading the electrochemical performance of ASSBs. Many studies have focused on identifying solvents that exhibit minimal reactivity toward sulfide electrolytes, such as acetonitrile, toluene, and tetrahydrofuran.^[37] Nonetheless, these solvents do not completely prevent the occurrence of side reactions. Additionally, efforts to mitigate these reactions through surface modifications such as doping or coating methods for sulfide-based electrolytes have often resulted in reduced Li^+ conductivity, limiting their practical application (Figure 3d).^[38] Conversely, the dry electrode process obviates the need for solvents, thereby avoiding side reactions with the solid electrolytes. This approach facilitates the fabrication of solid electrolyte membranes that retain ionic conductivities comparable to those of their powdered forms.^[39] Moreover, the shear forces applied during dry coating promote uniform dispersion of the electrode components, enabling the incorporation of various additives without dispersion constraints. In conclusion, beyond the degradation of the mechanical properties of the electrode caused by binder migration during the wet coating process, the necessity of the dry coating process becomes even more evident because of its impact on the electrode microstructure and chemical stability. Differences in binder morphology, the interface between the active materials and solid electrolytes, and the potential for side reactions with solvents further highlight the limitations of the wet coating process. By eliminating binder migration, the dry coating process not only improves the electrode microstructure and interfacial properties but also enhances chemical stability, ultimately leading to the superior electrochemical performance of ASSBs.

2.2. Dry Electrode Coating Process: Binders and Methods

The dry electrode process offers superior performance over the wet coating process in terms of both microstructural and chemical stability. Accordingly, various dry electrode coating techniques, along with suitable binders and solid electrolytes, have been extensively investigated. The mechanical properties of solid electrolytes vary depending on their chemical composition, which influences the application of dry electrode processing techniques (Figure 3e).^[40] For example, oxide-based electrolytes with a higher Young's modulus require high-temperature sintering to achieve densification, which poses additional processing challenges. In contrast, sulfide- and halide-based electrolytes with a low Young's modulus are easily deformable at room temperature, making them compatible with conventional calendaring processes used in electrode fabrication. The dry electrode process generally involves three main steps: powder mixing, free-standing electrode film fabrication, and current collector integration (lamination) (Figure 3f). During the powder mixing process, various electrode components are uniformly dispersed, whereas, during the free-standing electrode film fabrication, the dry mixture is compressed to form a film. Finally, during the lamination step, the fabricated electrode film is adhered to the current collector.^[41] Among the different roll-to-roll dry coating processes, the PTFE binder-based roll-to-roll dry coating process is the most actively studied.^[42–46] PTFE binders exhibit low phase transition temperatures (19 and 30 °C), and their crystal structures transform from triclinic to hexagonal above 19 °C, significantly reducing the storage modulus and allowing deformation under shear stress at room temperature, which does not occur in other conventional polymeric binders such as polyvinylidene fluoride (PVdF) and carboxymethylcellulose.^[47] The deformation of the PTFE binder under shear stress facilitates the formation of PTFE fibrous networks, enabling the fabrication of free-standing films at relatively low temperatures. Shin et al. fabricated high-nickel layered-structure cathodes using PTFE binders and investigated the relationship between the microstructures and electrochemical performance based on the composition variations of the electrode components.^[48] Currently, roll-to-roll dry coating processes that employ PTFE binders are broadly categorized into Maxwell and extrusion types (Figure 3g).^[49] Both the industry and academia are actively employing these methods to develop dry-processed electrodes and solid electrolyte membranes for ASSBs. Studies have explored electrostatic spraying and hot pressing methods using polyethylene oxide and PVdF binders, as well as fusion bonding techniques using thermoplastic polyamide (TPA) binders (Figure 3h,i).^[50–53] As PTFE is not perfect, research into alternative binders for the dry electrode coating process is ongoing. Kim et al. employed carboxylated nitrile butadiene rubber as a dry coating binder for sulfide-based solid electrolyte membranes, and Hong et al. developed poly(tetrafluoroethylene-co-perfluoro(3-oxa-4-pentenesulfonic acid)) Li salt, a Li^+ -conducting ionomer binder, for dry-processed cathodes, achieving high Li^+ conductivity (Figure 3j).^[54,55] Additionally, some studies have investigated binder-free systems that exploit the high ductility of sulfide-based electrolytes (Figure 3k).^[56] Despite these advancements, challenges remain in the commercialization of dry electrode processing techniques for ASSBs, necessitating further research and investment.

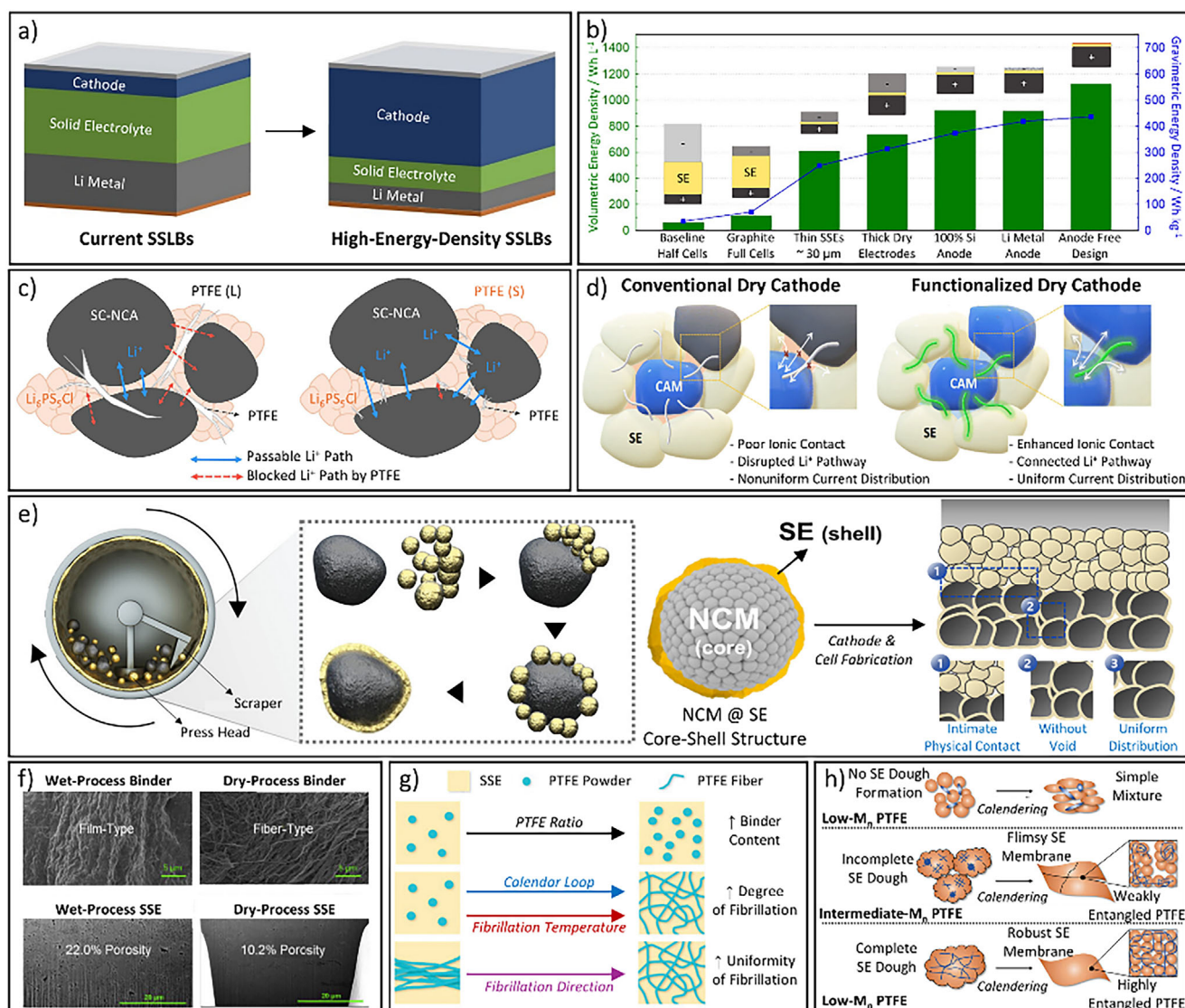


Figure 4. Strategies for high energy density ASSBs. a) Cell design and configuration for high-energy-density ASSBs and b) increase in energy density of ASSBs according to the various strategies. Reproduced with permission.^[58] Copyright 2022, OAE Publishing Inc. and ^[59] Copyright 2022, Elsevier. c) Control of the particle size of the PTFE binder, d) modification of the PTFE binder, and e) engineering of the interface between the cathode active materials and solid electrolyte for a thick dry cathode for ASSBs. Reproduced with permission.^[62] Copyright 2024, Elsevier B.V.,^[63] Copyright 2024, Royal Society of Chemistry and.^[64] Copyright 2023, Wiley-VCH. f) Differences in the porosity of solid electrolyte membranes prepared via the wet and dry coating processes. Reproduced with permission.^[59] Copyright 2022, Elsevier. g) Manipulation of the fibrillation behavior of the PTFE binder within the solid electrolyte membrane by controlling the processing parameters of the dry coating process. Reproduced with permission.^[65] Copyright 2023, Wiley-VCH. h) Investigation of the fibrillation behavior of the PTFE binder within the solid electrolyte membrane depending on its molecular weight. Reproduced with permission.^[66] Copyright 2024, Wiley-VCH.

2.3. High Energy Density ASSBs

Li anodes and anode-free systems have been widely explored to enhance the energy density of conventional LIBs. However, these approaches face significant challenges related to the safety issues associated with the presence of liquid electrolytes.^[57] In fact, ASSB systems have the advantage of employing Li anodes or anode-free systems. During their operation, applied external pressure can suppress the formation of Li dendrites, offering the potential for improved safety, electrochemical stability, and higher energy densities. However, beyond the adoption of

Li, silicon (Si) anodes, and anode-free systems, further advancements in cell design and configuration are required to achieve significantly higher energy densities in ASSBs. In this regard, the development of thick cathodes with a high cathode active material loading and ultrathin solid electrolyte membranes has been explored, which is possible through the introduction of the dry electrode process (Figure 4a,b).^[58,59] In LIB systems, the dry electrode process has already enabled the development of thick cathodes with low resistance, achieving areal capacities of $\approx 20 \text{ mAh cm}^{-2}$.^[60] However, when applying the dry electrode process technologies developed for LIBs to ASSBs, additional challenges

arise, particularly in controlling the interfaces between the solid electrolytes and cathode active materials to minimize void formation within the electrode and enhance the solid-state diffusion of Li^+ . Therefore, further technological advancements are required to address these challenges and optimize the dry electrode coating process for ASSB applications. Accordingly, material engineering approaches have been explored, such as controlling the particle size and crystallinity of the PTFE binder or incorporating additives to enhance Li^+ diffusion (Figure 4c,d).^[61–63] Moreover, Kim et al. developed core-shell structured electrolyte/cathode active material composites via a mechanofusion method, where sulfide-based solid electrolytes were coated onto the cathode active materials (Figure 4e).^[64] This approach effectively reduced the interfacial resistance and void formation, enabling the fabrication of thick, dry-processed cathodes with superior electrochemical performance.

The reduction in the thickness of the solid electrolyte membrane also contributed significantly to the increase in the energy density of ASSBs. Although wet coating processes have been employed to fabricate thin and dense electrolyte membranes, they often produce a solid electrolyte membrane with high porosity and a non-uniform microstructure due to the binder migration, which reduces the Li^+ conductivity of ASSBs (Figure 4f).^[59] Additionally, thin electrolyte membranes require a higher binder content to maintain their mechanical strength, which inversely affects the Li^+ conductivity of ASSBs. By continuously applying shear stress, the dry electrode coating process facilitates the fabrication of dense and thin solid electrolyte membranes with high mechanical strength, even at low binder contents. Accordingly, studies have focused on controlling the fibrillation behavior of the PTFE binder to obtain thin and mechanically robust solid electrolyte membranes. Lee et al. explored various fibrillation parameters, such as calendaring loops and mixing temperatures, to enhance the PTFE binder network within the solid electrolyte membrane (Figure 4g).^[65] Moreover, Yoon et al. developed sulfide-based solid electrolyte membranes with a low PTFE binder content of 0.5 wt%, achieving high mechanical robustness and a thickness of $\approx 13.5\ \mu\text{m}$ by utilizing high molecular weight PTFE binders that exhibited greater deformation and fibrillation (Figure 4h).^[66] These advancements in dry coating techniques have paved the way for the development of high-energy-density ASSBs with improved electrochemical performance.

3. Dry Electrode Process for Cathode Fabrication

Unlike the wet coating process, the dry electrode process eliminates the need for solvents by directly mixing the cathode material, conducting agent, and binder as powders. The mixture undergoes a kneading-like agglomeration process, followed by roll-pressing to form the electrode, which can then be integrated into LIBs. This solvent-free approach offers several advantages, including simplified processing, reduced environmental impact, and the potential for improved electrochemical performance. In this respect, in the next sections, we carefully review the recent progress in the development of dry-processed electrodes composed of $\text{LiNi}_x\text{Co}_y\text{Mn}_z\text{O}_2$ (NCM) and LiFePO_4 (LFP) cathode materials for practical applications in high-energy LIBs.

3.1. Roll-Pressing-Based Dry-Processed Cathode Electrode Fabrication

Ryu et al. investigated the electrochemical performance of a slurry-coated electrode (SCE) and a dry press-coated electrode (DPCE) for LIBs. In the dry electrode processing cathode, 1-D multiwalled carbon nanotubes (MWNTs) were employed as conductive additives instead of 0-D Super-P. DPCE was fabricated by hot-pressing and roll-pressing a blend of $\text{LiNi}_{0.7}\text{Co}_{0.1}\text{Mn}_{0.2}\text{O}_2$ (NCM712), PVdF, and MWNTs onto Al foil. DPCE exhibited a superior rate capability and cycling performance compared with SCE (Figure 5a).^[51] At a current density of 1.0 C, DPCE maintained a higher discharge capacity of $170\ \text{mAh g}^{-1}$ and much higher capacity retention of 67% after 400 cycles, compared with SCE ($159\ \text{mAh g}^{-1}$ with 35% retention). In high-loading configurations, DPCE (HL-DPCE) was tested in pouch half-cells with a Li metal anode, achieving an areal capacity of $17.6\ \text{mAh cm}^{-2}$ at a loading of $100\ \text{mg cm}^{-2}$. The specific and volumetric energy densities of the resulting LIBs were estimated to be $360\ \text{Wh kg}^{-1}$ and $701\ \text{Wh L}^{-1}$, respectively (Table 1). Kwon et al. also investigated dry electrode-processed cathodes using LFP by comparing their electrochemical performance with that of wet-processed LFP cathodes (Figure 5b).^[43] A wet-processed cathode was fabricated by mixing LFP, PVdF, and carbon black (Super-P) in an NMP solvent, whereas a dry-processed cathode was prepared using LFP, PTFE, and Super-P without any solvent. The electrochemical performance of both cathodes was compared at a loading of $5.5\ \text{mAh cm}^{-2}$. In half-cell configurations, the wet-processed cathode exhibited a higher overpotential and resistance, as well as a lower Li^+ ion diffusion coefficient than the dry-processed cathode, showing a significant decrease in its areal capacity at a high current density of 1 C. Furthermore, in full-cell configurations (anode: graphite), the dry-processed cathode demonstrated superior rate capability. At an increased loading of $7.8\ \text{mAh cm}^{-2}$ the dry-processed cathode maintained its excellent areal capacity and capacity retention over 300 cycles (Table 1). However, these studies also reveal key challenges, such as increased polarization effects and Li^+ transport limitations due to higher tortuosity. The performance trade-offs observed at higher C-rates underscore the need for further optimization, particularly in terms of structural design and Li^+ transport pathways.

3.2. Alternative Binder (Pitch) and Binder-Less Dry-Processed Cathode Electrodes

Park et al. evaluated the electrochemical performance of high-loading cathodes for LIBs fabricated using the wet and dry processes with $\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$ (NCM622) as the cathode material and Super-P as the conducting agent. In the wet process, PVdF was used as the binder, whereas pitch was employed in the dry process (Figure 5c).^[67] The wet process involves four steps to produce the NCM622 cathode, whereas the dry electrode process requires only two steps, enabling the fabrication of thicker cathodes. The maximum thickness achieved for the wet-processed cathode was $150\ \mu\text{m}$, while that of the dry-processed cathode was $300\ \mu\text{m}$, effectively doubling the cathode thickness and demonstrating its potential for high-energy LIB applications. Electrochemical performance comparisons revealed that at

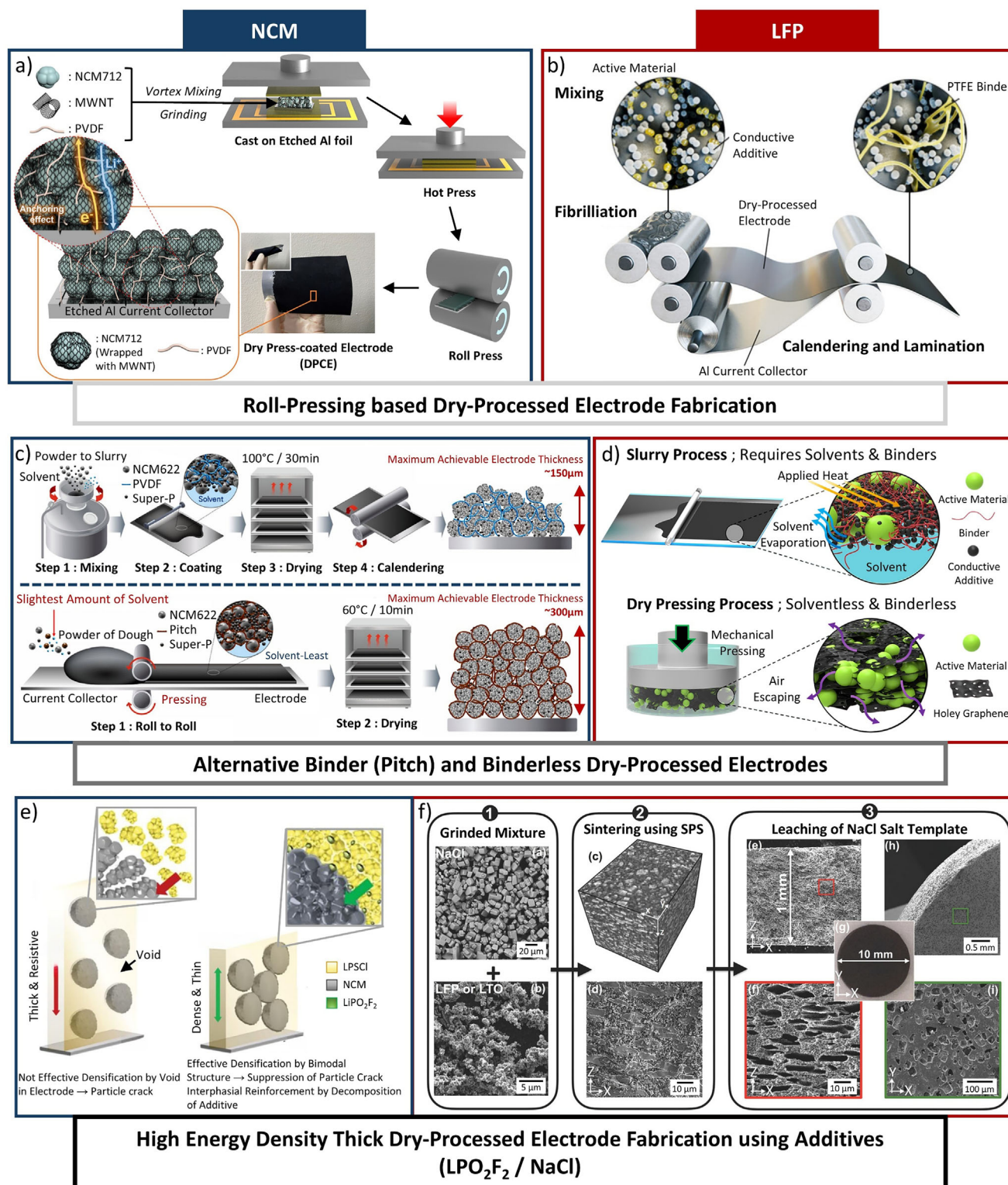


Figure 5. a) Schematic of the fabrication procedure and structural design of DPCE. Reproduced with permission.^[51] Copyright 2023, Springer Nature. b) Schematic of the dry electrode processes of the LFP cathode active material. Reproduced with permission.^[43] Copyright 2024, Wiley. c) Schematic of the fabrication procedure and structural design of the WP thick electrode using PVdF as a binder and the DLS thick electrode using pitch as a binder. Reproduced with permission.^[67] Copyright 2024, Elsevier B.V. d) Overview of the fabrication process of the slurry-based electrode with a binder and the dry compression process with hG. Reproduced with permission.^[68] Copyright 2019, American Chemical Society. e) Schematic of the distinctive effectiveness of LiPO₂F₂ as an electrode additive for ASSBs. Reproduced with permission.^[70] Copyright 2024, Wiley-VCH. f) The three steps involved in the fabrication of thick porous electrodes using NaCl particles as templating agents. Reproduced with permission.^[71] Copyright 2018, Wiley-VCH.

Table 1. A summary of the cell configurations, compositions, and electrochemical performance of the various battery systems is presented in Figure 5.

Cell type	Electrode composition [wt%]				Thickness [μm]	Mass loading [mg cm ⁻²]	Areal capacity [mAh cm ⁻²]	Discharge current density [mA cm ⁻²]	Reversible specific capacity [mAh g ⁻¹]	Cycle retention [%@cycle]	Energy density [Wh kg ⁻¹ /Wh L ⁻¹]
	AM ¹	LE ² /SE ³	CA ⁴	Binder							
Figure 5a Pouch cell (LIB)	80(NCM712)	1 M LiPF ₆ in EC/DEC 1/1 (w/w)	15(MWNTs)	5(PVdF)	573	100	17.6	1.76	176	67.0%@400th cycle	360 / 701
Figure 5b Pouch cell (LIB)	97(LFP)	1.15 M LiPF ₆ in EC/EMC/DMC 2/4/4 (v/v/v)	1(Super-P)	2(PTEF)	175	52	7.8	2.4	154.6	65.5%@300th cycle	185 / 470
Figure 5c 2016R coin-type cell (LIB)	90(NCM622)	1.3 M LiPF ₆ in EC/EMC/DMC 3/5/2 (v/v/v) with various additives	7(Super-P)	3(Pitch)	305.1	75.1	10.9	1.09	170.5	85.5%@15th cycle	–
Figure 5d CR2032 coin-type cell (LIB)	50(LFP)	1 M LiPF ₆ in EC/EMC 3/7 (v/v)	50(hG)	–	340	11.6	0.9	0.18	> 160	87.5%@200th cycle	–
Figure 5e Pressure cell (ASSB)	85(NCM811)	13 (LPSCI)	1(Super-C)	1(PTEF)	156.8	57.5	10.0	0.5	208.86	96.5%@100th cycle	–
Figure 5f classical Swagelok cells (LIB)	50(LFP)	1 M LiPF ₆ in EC/DMC 1/1 (v/v)	10(Super-P)	–	1000	114.3	20	1	178	84%@10th cycle	–

AM¹: Active material; LE²: liquid electrolyte; SE³: solid-state electrolyte; CA⁴: conducting agent.

loadings below 10.0 mg cm^{-2} , the wet-processed cathode exhibited a higher discharge capacity of 180.0 mAh g^{-1} compared with the 170.5 mAh g^{-1} of the dry-processed cathode. However, at higher loadings, the dry coating processed cathode showed superior areal capacity (10.9 mAh cm^{-2}) with a high initial coulombic efficiency of 96.4%, highlighting its advantages for increasing the energy density of LIBs (Table 1). Kirsch et al. fabricated binderless and solvent-free dry-processed cathodes using holey graphene (hG) as the conductive additive for a practical comparison with conventional wet-processed cathodes (Figure 5d).^[68] The wet-processed cathode was prepared using a standard slurry-based method, whereas the dry-coating-processed cathode was fabricated by compressing a mixture of LFP and hG powders. During compression, air was expelled from the powder mixture to form dense binder-free and solvent-free cathodes. The thickness of the dry cathodes varied depending on the applied pressure, being $160 \text{ }\mu\text{m}$ at 500 MPa, $175 \text{ }\mu\text{m}$ at 200 MPa, and $340 \text{ }\mu\text{m}$ at 20 MPa. Even at a low pressure of 20 MPa, the dry-processed cathode maintained its structural integrity, with LFP and hG remaining uniformly distributed. The dry-processed cathode exhibited excellent capacity retention over 200 cycles. Moreover, the dry-processed cathode demonstrated a discharge capacity of $\approx 70 \text{ mAh g}^{-1}$ at a high current density of 3 C, confirming its superior rate performance (Table 1).

3.3. High Energy Density Thick Dry-Processed Cathode Electrode Fabrication Using Additives

Recently, Kim et al. optimized the dry electrode process for cathode fabrication using $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (Ni-rich, NCM811), PTFE, and either Super-C or carbon nanofibers (CF) as conductive additives.^[69] The dry-processed cathodes with CF showed a linear contact, reducing performance degradation in ASSBs and achieving a high areal capacity of $\approx 6 \text{ mAh cm}^{-2}$ with over 98% retention after 150 cycles. Further improvements were reported by incorporating LiPO_2F_2 into the dry-electrode process to reduce the thickness while maintaining a high loading due to the pore-filling effect (Figure 5e).^[70] This additive enhanced electrode density, resulting in a higher volumetric capacity of 600 mAh cm^{-3} compared with the 500 mAh cm^{-3} obtained without the additive (Table 1). Elango et al. developed ultra-thick, binder-free dry-processed cathodes ($\approx 20 \text{ mAh cm}^{-2}$) using the spark plasma sintering (SPS) method, achieving four times the areal capacity of conventional $100\text{-}\mu\text{m}$ -thick tape-cast cathodes ($\approx 5 \text{ mAh cm}^{-2}$). Instead of a binder, micro-sized NaCl was used as the templating agent. NaCl particles were mixed with LFP or $\text{Li}_4\text{Ti}_5\text{O}_{12}$ (LTO) before sintering via SPS, leading to the successful fabrication of a $\approx 1\text{-mm}$ -thick dry-processed cathode (Figure 5f).^[71] This binder-free cathode exhibited a high porosity of 40%, with uniformly distributed pores that facilitated electrolyte penetration. A full cell was assembled using an LFP cathode and an LTO anode, achieving an areal capacity of 20 mAh cm^{-2} and demonstrating stable capacity retention up to 20 cycles. Furthermore, tortuosity measurements along the x, y, and z axes for the dry-processed cathodes indicated a higher tortuosity along the z-axis, suggesting that pore formation during SPS may impact Li^+ ion transport. However, further research is required to optimize the struc-

tural design of thick, dry-processed cathodes to mitigate these Li^+ transport limitations (Table 1).

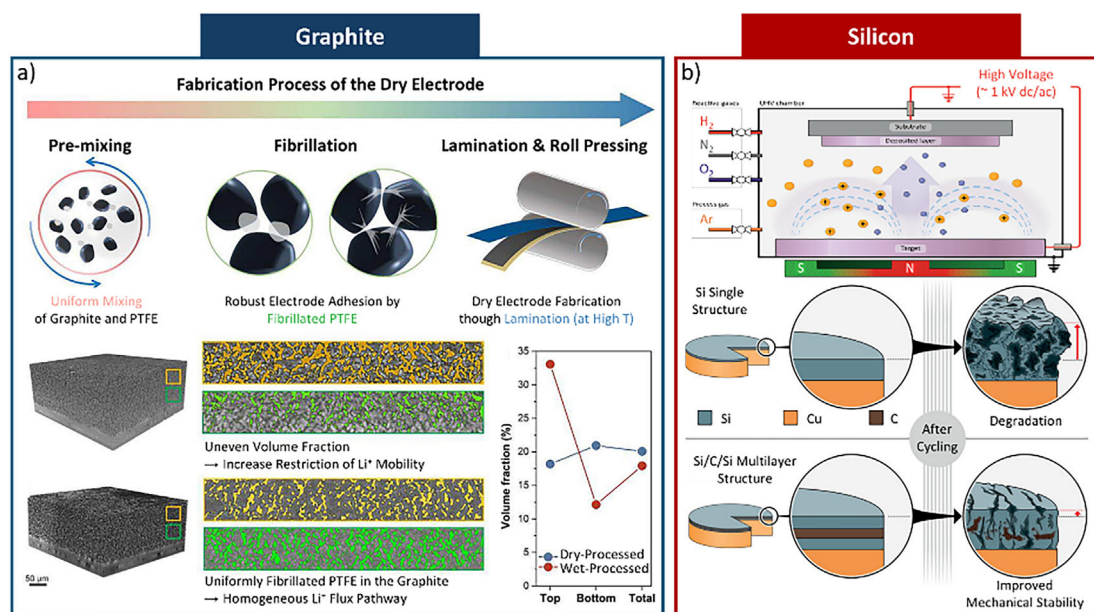
The dry electrode process for cathode fabrication simplifies production, reduces costs, and eliminates the need for solvents, enabling the production of high-loading cathodes. By integrating various binders and conductive and functional additives, the dry electrode-coating process for the cathodes affords improved areal capacity, volumetric capacity, and cycle retention. These advancements highlight the feasibility of developing high-energy LIBs and ASSBs to accelerate their commercialization.

4. Dry Electrode Process for Anode Fabrication

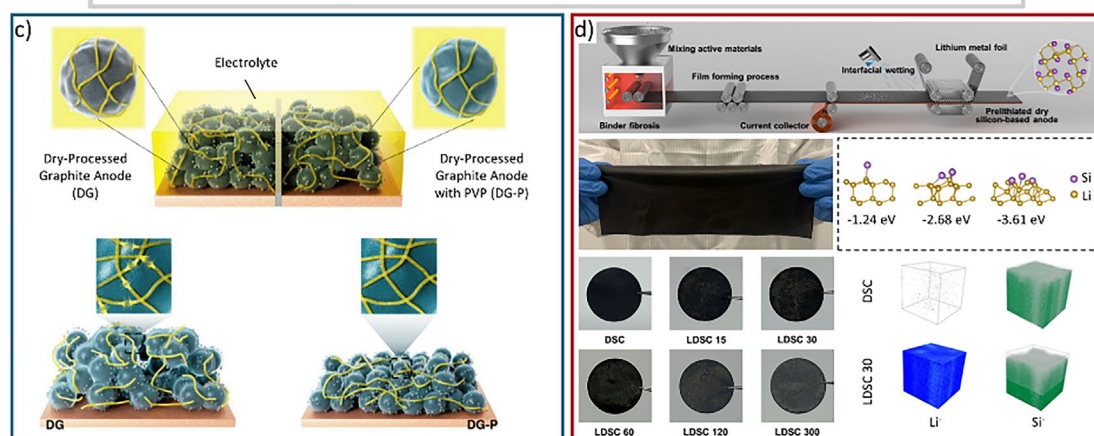
To develop high-energy LIBs, both the cathode and anode must be designed carefully to have high capacity and potential gap.^[72] To achieve this, recent studies have focused on developing thick, dry-coated anodes. Although graphite has traditionally been used as the anode active material in dry electrodes, its low capacity has led to the exploration of alternative active materials. Si-based dry-processed anodes and anode-less designs have emerged as promising next-generation solutions, with the Ag-C coating layers facilitating Li metal deposition.

4.1. Fabrication of Dry-Processed Anode Electrodes Using Various Processing Techniques

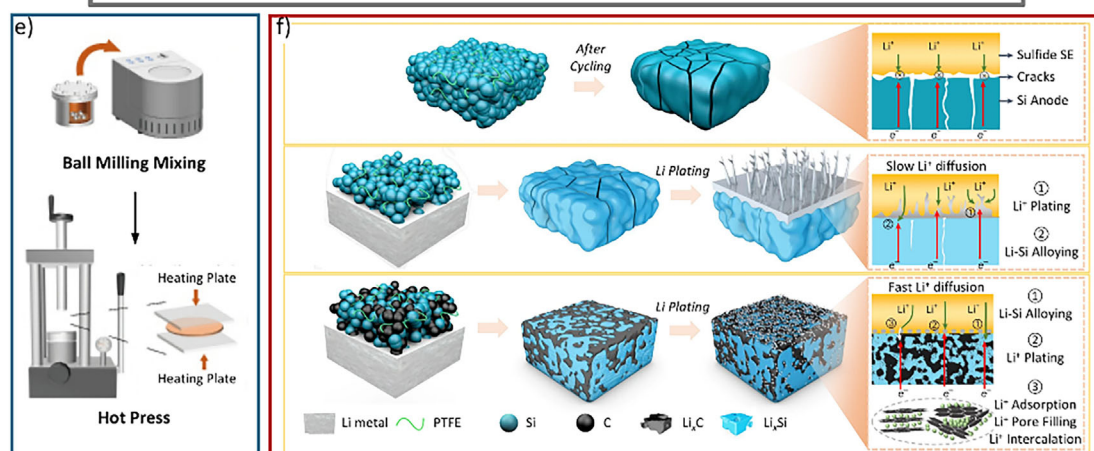
Suh et al. developed a dry-processed anode by mixing graphite with the PTFE binder and compared its performance with that of a conventional wet-processed anode. The dry anode fabrication process involves three main steps: pre-mixing, fibrillation, and lamination with roll pressing. The resulting dry-processed anode, with an areal capacity of 6 mAh cm^{-2} , was analyzed using 3D digital imaging and X-ray microscopy (3D-XRM), and its pore distribution was compared with that of the wet-processed anode. The wet-processed anode exhibited a higher porosity at the top and lower porosity at the bottom, whereas the dry-processed anode exhibited a more uniform pore distribution (Figure 6a).^[73] This difference in porosity likely contributed to the variations in internal resistance between the two electrode types. While both anodes showed similar capacities during formation at 0.1 C, resistance measurements at different states of charge revealed that the dry-processed anode had a lower resistance, suggesting that pore distribution plays a critical role in influencing the resistance of LIBs. The full cells assembled with NCM622 cathodes were tested at 0.5 C. The dry coating-processed anode demonstrated higher capacity retention after 300 cycles, and voltage profile analysis further confirmed its lower internal resistance, highlighting the advantages of dry processing for long-term cycling stability (Table 2). Reyes et al. explored a solvent-free dry-electrode fabrication method using magnetron sputtering to deposit multiple layers of Si and carbon (C). This approach aimed to mitigate the degradation issues associated with single Si structures that suffer from significant volume expansion and structural instability during cycling. Unlike the conventional roll-press dry electrode fabrication method, this method relies on magnetron sputtering to form a layered Si/C electrode structure without a binder. The resulting Si/C/Si multilayer structure was directly infiltrated with a



Fabrication of Dry-Processed Electrodes using Various Processing Techniques



Surface Modification of Active Materials to Suppress Side Reactions in Dry-Processed Electrodes



Fabrication of Fast Charging Dry-Processed Electrodes using Electrically Conductive Materials (PANI / Hard Carbon)

liquid electrolyte, enabling electrochemical operation without the need for the conventional slurry-based processing (Figure 6d).^[74] During electrochemical testing, the Si/C/Si dry-processed electrode exhibited a discharge capacity of $\approx 250 \text{ mAh g}^{-1}$ and retained 83% of its initial capacity after 150 cycles, demonstrating improved cycling stability compared with the single Si structure. This study highlights the potential of magnetron sputtering as an alternative dry fabrication technique for next-generation LIB electrodes, particularly Si-based anodes (Table 2).

4.2. Surface Modification of Active Materials to Suppress Side Reactions in Dry Coating Processed Anode Electrodes

Lee et al. addressed the technical issue of PTFE decomposition in dry-processed anodes by incorporating polyvinylpyrrolidone (PVP). PVP was first kneaded with graphite under shear force before anode fabrication. A protective PVP layer was formed on the graphite surface, which prevented direct contact between the graphite and PTFE, thereby improving the Li^+ kinetics and surface stability by forming a robust inorganic-rich solid electrolyte interphase (SEI). Figure 6 presents the structural differences between the dry-processed anodes without PVP (DG) and those with PVP (DG-P), with DG-P significantly reducing the PTFE decomposition (Figure 6c).^[75] The electrode thickness decreased from 112 to 98 μm after the first cycle, indicating improved densification. At a high loading level ($\approx 10 \text{ mAh cm}^{-2}$), the discharge capacity reached 9.3 mAh cm^{-2} , with a cycle retention of 86.3% after 200 cycles, demonstrating the enhanced durability and performance of the dry-processed anodes (Table 2). Dong et al. investigated the feasibility of fabricating dry-processed anodes using Si, a promising anode material for next-generation high-energy LIBs. A dry Si-carbon-based anode (DSC) was fabricated using Si, Super-P, and PTFE. To induce pre-lithiation, a liquid electrolyte (1 M LiPF_6 in a 1:1 (v/v) mixture of ethylene carbonate and diethyl carbonate containing 1% vinyl carbonate and 10% fluoroethylene carbonate) was sprayed onto DSC, followed by contact with Li to form a lithiated DSC (LDSC). The samples were labeled LDSC 15–300 based on the lithiation time, and a surface morphology analysis was conducted. Time-of-flight secondary ion mass spectrometry (ToF-SIMS) analyses confirmed the uniform Li^+ distribution in LDSC-30 (Figure 6d),^[76] which was fabricated as a dry-processed electrode with a thickness of 70.21 μm . During full-cell evaluation, the electrode demonstrated a discharge capacity of 175.7 mAh g^{-1} and maintained a high cycle retention of 88.2% after 300 cycles (Table 2). These results confirmed the improved cycling stability and long-term durability of the LDSC-30 electrode and demonstrated the feasibility of using LDSC anodes,

highlighting the benefits of pre-lithiation for enhanced electrochemical performance.

4.3. Fabrication of Fast Charging Dry Coating Processed Anode Electrodes Using Electrically Conductive Materials (Polyaniline/Hard Carbon)

Yuan et al. investigated polyaniline (PANI) as a dual-functional material serving as both a binder and a conductive additive, replacing the conventional PVdF binder. Carbon-based anode materials have high electronic conductivities; therefore, additional carbon additives are typically unnecessary. However, substituting the nonconductive PVdF binder with conductive PANI could enhance the rate capability and cycle performance. A dry-processed anode was fabricated using graphite and PANI via hot pressing (Figure 6e).^[77] The dry-processed anodes with PANI and PVdF showed similar initial formation capacities; however, after 100 cycles, the PANI-based anode exhibited superior capacity retention (78.2% @ 100th cycle) (Table 2). Yan et al. developed lithiated Si-hard carbon hybrid (LiSH) anodes by incorporating Si, Li, and hard carbon (HC) into dry-processed anodes to achieve high-energy and high-power ASSBs. Various Si-to-HC ratios were tested, resulting in compositions including LiSi, LiSH82, LiSH64, LiSH46, LiSH28, and LiHC. Optimization studies and comparison of the electrochemical performances identified LiSH46 as the optimal composition (Figure 6f).^[78] Subsequently, a high-loading ($\approx 6 \text{ mAh cm}^{-2}$) dry-processed anode demonstrated a cycle retention of 61.5% after 5000 cycles at 1 C with excellent rate capability in ASSBs (Table 2).

To achieve high-energy LIBs, it is essential to increase the areal capacity of dry-processed anodes. Similar to cathodes, anodes are fabricated by mixing active materials with binders to achieve uniform pore distribution. However, unlike cathodes, anodes operate at lower voltages, making PTFE less suitable because of its instability under these conditions.^[79,80] To address this limitation, researchers are exploring alternative binders and additives to suppress PTFE decomposition and improve anode performance. Despite these advancements, the low theoretical capacity of graphite (372 mAh g^{-1}) remains a major barrier to achieving high energy density in LIBs. Therefore, further research is required to identify the next-generation anode materials that can replace graphite and support the development of high-energy LIBs. Si is being widely investigated as a replacement for graphite in LIB anodes owing to its significantly higher theoretical capacity. While graphite has a theoretical capacity of 372 mAh g^{-1} , Si offers a much higher capacity of $\approx 3579 \text{ mAh g}^{-1}$, making it a promising candidate for next-generation high-energy-density

Figure 6. a) Schematic of the dry-electrode process for fabricating graphite anodes, along with 3D XRM reconstructed images of the wet- and dry-processed electrodes. The cross-sectional images show the pore distribution in the top and bottom regions and a comparison of the porosity in the top, bottom, and overall layers of the wet- and dry-processed electrodes. Reproduced with permission.^[73] Copyright 2023, Elsevier B.V. b) Schematic of the magnetron sputtering process and the resulting Si/C and Si/C/Si thin-film electrodes. Reproduced with permission.^[74] Copyright 2017, American Chemical Society. c) Schematic of the dry-processed graphite anode (DG) and the dry-processed graphite anode with PVP (DG-P); comparison of the DG and DG-P performances after cycling. Reproduced with permission.^[75] Copyright 2025, Elsevier B.V. d) Schematic of the pre-lithiation process for the dry electrode and a photograph of the obtained large-scale LDSC 30 anode, including DSC and LDSC anodes with various pre-lithiation reaction times. Also shown are the calculated binding energies of the various Li_xSi phases and TOF-SIMS 3D reconstruction images of the Li and Si distributions in DSC and LDSC. Reproduced with permission.^[76] Copyright 2025, Elsevier B.V. e) Flow diagram of solvent-free dry graphite anode production. Reproduced with permission.^[77] Copyright 2024, Elsevier Ltd. f) Schematic of the operating mechanisms for the Si, LiSi, and LiSH46 anodes in ASSBs. Reproduced with permission.^[78] Copyright 2023, Springer Nature.

batteries. However, the fabrication of dry-processed anodes using Si requires additional processing steps compared with conventional graphite-based anodes. Because Si consumes a significant amount of Li^+ during SEI formation, leading to a high irreversible capacity loss, pre-lithiation is applied post-fabrication to pre-form the SEI layer and mitigate capacity loss.^[81,82] However, even with pre-lithiation, pure Si anodes exhibit challenges. Therefore, Si is typically blended with carbon-based materials to leverage the complementary properties of graphite and Si, which results in enhanced performance over that of single-component anodes.

5. Auxiliary Components

Rechargeable batteries incorporate various supporting materials to facilitate the electrochemical reactions of the active materials, ensure current flow, and deliver reliable performance. Regarding ASSBs, the distinctive characteristics of solid electrolytes require a specific understanding of these auxiliary components. In this section, particular attention is paid to the current collectors and conductive additives, with a focus on their roles in the fabrication of dry electrodes for ASSBs.

5.1. Solid Electrolyte Membrane

For ASSBs, the solid electrolyte sheet must possess reliable characteristics to facilitate high ionic conductivity while effectively blocking electronic conduction. Achieving rapid and massive ionic transportation of the solid electrolyte sheet is critical to minimizing the internal resistance of ASSBs. The dry electrode process, which relieves surface degradation of fragile solid electrolytes, is fundamentally advantageous for mitigating surface resistance. Furthermore, the free-standing nature of dry-processed electrolyte sheets is correspondent with the design requirements for ASSB applications. The ductile properties of sulfide-based solid electrolytes are particularly advantageous, as they facilitate the reduction of defect pores during dry-shearing processes, outperforming sheets produced by conventional wet processes involving solvent drying.^[31]

Despite the intrinsic safety advantages offered by ASSBs, several recent papers have highlighted safety concerns in ASSBs. Many of these issues stem from Li dendrite formation, which can lead to internal short circuits between the cathode and anode. The use of robust electrolyte sheets can block such failure modes. However, inhomogeneities in conductivity or localized high resistance within the electrolyte sheet can result in uneven current flux and non-uniform Li plating at the anode interface. To support high ionic conductivity in electrolyte sheets, it is essential to ensure dense physical contact among solid electrolyte particles, regulate grain boundary resistance, and sustain a homogeneous distribution of binder to advance the structural integrity of the sheet. Accordingly, the design of binder distribution and mixing in the dry electrode process must be carefully optimized. A higher binder content improves physical stability and particle connectivity; however, the insulating nature of polymeric binders negatively impacts ionic conductivity. In this regard, dry-processed solid electrolyte sheets incorporating minimal PTFE binder, as low as 0.1 wt.%, have been demonstrated.^[33]

The fibrillation of PTFE during the dry process enables the formation of mechanically robust, free-standing sheets even with low binder content. In addition, the implementation of interlayers has emerged as a key strategy for enhancing the performance of high-energy-density anode systems, such as Li-free configurations and Li metal.^[6,7] These interlayers are highly effective in stabilizing the solid electrolyte/anode interface, suppressing dendrite growth, and facilitating uniform Li deposition, thereby significantly improving the electrochemical stability and cycle life of ASSBs.^[83,84]

New types of binders for dry-processed solid electrolyte sheets are under investigation to overcome the limitations of PTFE. From the perspective of electrochemical integrity, PTFE is not an ideal solution.^[44,79] When combined with unstable sulfide solid electrolytes, PTFE can react to form by-products such as LiF and organic species, which may introduce electronic pathways and trigger internal short circuits.^[65] This failure mode is accelerated by more fibrillization to cause long-connected by-products in the electrolyte sheets. Nevertheless, PTFE-based sheets remain viable candidates for solid electrolyte fabrication. High-molecular-weight PTFE has shown enhanced electrochemical stability, albeit at the expense of reduced mechanical flexibility. Supporting agents such as ionic liquids and organic solvents have been employed to further lower surface resistance.^[85,86] Efforts to refine the dry electrode process continue. Solid electrolyte sheet fabrication process with high-temperature press supports homogeneous binder distributions with thermally flexible characteristics of PTFE. Notably, Kim et al. reported a cryo-pulverization method for PTFE binders, which improves binder distribution from hundreds of micrometers to tens of micrometers while promoting fibrillation and preserving surface integrity.^[62] Alternative binders with vitrimers,^[87] styrene-butadiene rubber (SBR),^[88] TPA^[52] have been reported through modified dry electrode processes as well.

5.2. Current Collectors

Current collectors serve as electronic conductors that connect the electrode terminals to external circuits, facilitating the charge distribution over the electrodes during electrochemical reactions. Usually, current collectors are conductive metal foils or mesh-composed of electrochemically inactive materials. Suppressing undesired electrochemical reactions at the current collectors is essential because they can induce pulverization or the formation of insulating phases, thereby compromising the overall battery performance.

In ASSBs, aluminum or stainless steel is commonly used as the cathode current collector, whereas materials such as Ni, Fe, and stainless steel are employed as the anode.^[89,90] Conventional copper (Cu) current collectors, which are the most common current collectors in LIBs with liquid electrolytes, have demonstrated instability in sulfide-based ASSBs. The formation of copper sulfides due to reactions with H_2S generated from sulfide electrolytes increases interfacial resistance. Given that metal sulfides typically exhibit low Gibbs free energies of formation, their spontaneous formation is thermodynamically feasible. Proper selection of metal foils with native kinetic barriers such as surface metal oxide layers can inhibit sulfidation degradation. However,

Table 2. Summary of the cell configurations, compositions, and electrochemical performances of the various battery systems presented in Figure 6.

Cell type	Electrode composition [wt%]				Thickness [μm]	Mass loading [mg cm ⁻²]	Areal capacity [mAh cm ⁻²]	Discharge current density [mA cm ⁻²]	Reversible specific capacity [mAh g ⁻¹]	Cycle retention [%@cycle]	Energy density [Wh kg ⁻¹ /Wh L ⁻¹]
	AM ¹	LE ² /SE ³	CA ⁴	Binder							
Figure 6a CR2032 coin-type cell (LIB)	98(Graphite)	1.15 M LiPF ₆ in EC/EMC/DMC (25/45/30) (v/v/v) with various additives	–	2(PTFE)	–	41.25(Anode)	6	2.75	160(Cathode)	88.2%@300th cycle	–
Figure 6b CR2032 coin-type cell (LIB)	73.7(Si)	1 M LiPF ₆ in EC/DEC (3/7) (w/w) with 2 wt% VC	26.3(C)	–	0.2	–	–	–	≈250(Cathode)	83%@150th cycle	–
Figure 6c CR2032 coin-type cell (LIB)	96(Graphite)	1.15 M LiPF ₆ in EC/EMC/DEC (2/4/4) (v/v/v) with various additives	1(Super-P)	3(PTFE)	0.5	93 (pristine) (PVP)	30.2(Anode)	1.8	341.5(Anode)	86.3%@200th cycle	–
Figure 6d CR2032 coin-type cell (LIB)	96(Si/C)	1 M LiPF ₆ in EC/DEC (1/1) (v/v) with various additives	1(Super-P)	0.8(PTFE)	–	70.21	–	–	175.7(Cathode)	88.2%@300th cycle	–
Figure 6e CR2032 coin-type cell (LIB)	80(Graphite)	–	–	20(PANI)	–	–	2.67	1.03	221.4(Anode)	78.2%@100th cycle	–
Figure 6f Pressure cell (ASSB)	95(Si)	–	–	5(PTFE)	(HC)	44.7	2.2	5.86	2803.8(Anode)	61.5%@5000th cycle	–

AM¹: Active material; LE²: liquid electrolyte; SE³: solid-state electrolyte; CA⁴: conducting agent.

elevated temperatures accelerate thermodynamic reactions, posing challenges to the stability of current collectors.^[91] From this perspective, compared with wet methods, shorter heat treatment durations in dry electrode processing offer greater flexibility in the selection of current collectors in terms of chemical side reactions. Furthermore, the absence of a solvent for electrode fabrication reduces the possibility of H₂S gas emission, which is harmful as well as corrosive to current metal collectors.

Various functional designs have been reported for current collectors to enhance the contact area with active materials and reduce the ohmic resistance. This has been approached in various ways, from controlling the roughness of the entire surface of the material to utilizing efficient novel 3D structures.^[92,93] In the dry electrode process, the electrode mixture has low flowability and is attached to the current collector under pressure during the final stage of fabrication. Thus, the variety of current collector morphologies is usually limited compared with that in the wet electrode process.

The lamination process requires strong adhesion to the current collector through interactions between the binder and collector metal. During the charge and discharge processes of a cell, the active material undergoes volume changes because of the insertion and extraction of Li⁺. Although this induces physical stress within the electrode itself, the current collector, which does not participate in the electrochemical reactions, does not experience such volume changes. As a result, the interfacial region between the electrode and current collector is subjected to amplified mechanical stress. Therefore, a strong interfacial adhesion between the dry-coated electrode and the current collector is essential to ensure mechanical integrity and long-term stability. In this context, mesh-type current collectors, which are uniquely applicable to dry electrode systems, can be effectively utilized to enhance interfacial adhesion.

5.3. Conducting Agents

Conducting agents (or conductive additives) are usually nano-sized carbon powders dispersed in the electrodes to ensure a high electronic conductivity. In wet-slurry electrode processing, lowering the slurry viscosity enhances flowability and processability; however, it may lead to the agglomeration of conducting agent particles, thereby limiting their dispersion. Dry electrodes circumvent these re-agglomeration drawbacks. In sulfide-based ASSBs, wet electrode processing suffers from interfacial stability issues between the solvents and solid electrolytes, necessitating shorter slurry processing times. Such a constraint impedes the uniform dispersion of the conducting agents. Furthermore, the high volatility of some electrolyte solvents can induce bubble formation and nonuniform distribution of conductive nanopowders and binders during drying.

Dry electrode processing bypasses these issues, allowing greater control over the dispersion of conductive additives. The absence of drying steps ensures uniform electrode characteristics, rendering dry-electrode processing a highly promising technology for next-generation ASSBs.

Low-surface-area 1- or 2D conducting agents exhibit high conductivities owing to their high crystallinities. To achieve a high rate capability in ASSBs, a balance between the ionic and elec-

tronic conductivities is required. Although carbon enhances electronic conductivity, excessive amounts of carbon negatively affect both the energy density and ionic conductivity.^[94] Highly conductive multidimensional carbons are beneficial for reducing the Cobalt content in electrodes; however, their poor dispersibility remains a drawback.^[2,94–99] The dry electrode process can enhance the variety and accessibility of carbon materials while overcoming mixing time constraints, resulting in well-distributed conducting agents for ASSB electrodes.

6. Perspective

The establishment of a dry electrode fabrication process for ASSBs is important to realize their inherent advantages. Considering global warming and the intense competition in the battery industry, the use of dry ASSB electrodes is inevitable. However, for the wide application of the ASSB dry electrode processes, several challenges must first be addressed (Figure 7).

The dry electrode process requires the development of specialized component materials, especially novel binder substances, to address the challenges associated with the high ionic resistance of the binder. During the dry coating process, the PTFE binders undergo fibrillar action under a shear force, forming a fibrous network structure that enhances the mechanical strength within the electrode. However, the PTFE binders exhibit a low ionic conductivity, which may hinder Li⁺ transport within the electrode. To address this limitation, composite binders incorporating ionically conductive binders (e.g., Li-ionomer) along with PTFE binders should be considered. Although previous studies have explored the use of sulfide-based solid electrolytes without binders because of their high ductility, the feasibility of employing binder-free systems for large-scale manufacturing remains limited. Therefore, further research into the simultaneous application of composite binder systems is necessary to achieve mechanical stability and enhanced ionic conductivity. PTFE undergoes irreversible electrochemical reactions with Li⁺ at low potentials and chemical reactions with sulfide electrolytes. The development of novel polymer binder with improved electrochemical and chemical stability is highly desirable. Building polymers with tailored architectures and optimizing polymer combinations can significantly relieve the ASSB dry electrode process.

The dry electrode fabrication procedure can effectively mitigate issues such as conductive additive aggregation, which often occurs in conventional slurry-based processes due to the low viscosity of the slurry. Efficient electrochemical reactions require well-established pathways for both electron and ion transport. This can be achieved by ensuring a uniform distribution of the solid electrolyte and conductive additives throughout the electrode. However, the dispersion of conductive additives in dry coating systems remains inherently more challenging. Without the aid of solvent-driven dispersion mechanisms, particle–particle interactions dominate the dry system, often resulting in localized agglomeration—particularly for conductive additives with high aspect ratios or nanoscale dimensions. Although point-to-point contact-based binders are typically employed, allowing for the use of various nano-morphological conductive additives, these do not inherently ensure uniform distribution throughout the dry electrode. The fibrillization of the PTFE binder under shear force helps to entangle conductive additives within a fibrous

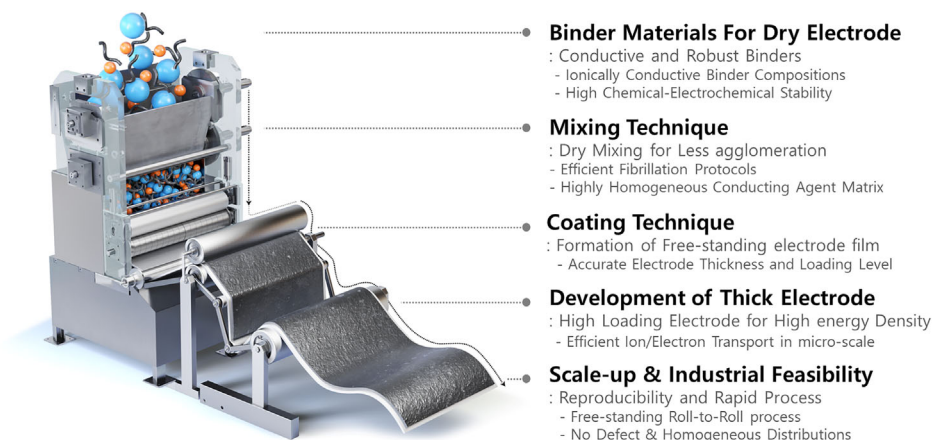


Figure 7. Schematic of the perspective of the use of dry electrode process for ASSBs with the relevant challenges.

network, thereby contributing to a more homogeneous electrode microstructure. However, this mechanism alone is insufficient to ensure the formation of a continuous and efficient electronic conduction network. While such microstructural uniformity offers opportunities for optimizing the electron-ion transport balance in ASSBs, a critical gap remains in achieving a robust, percolated electronic network—particularly in thick electrodes. In contrast to the wet coating process, which utilizes solvents and dispersants to facilitate the uniform distribution of conducting agents, the dry coating process currently lacks a well-established strategy for dispersion control. This highlights the urgent need for novel solid-state dispersants that are chemically compatible, mechanically robust, and capable of suppressing conductive additive agglomeration. In wet mixing processes, the binder remains largely unaffected by mixing time and shear stress. In contrast, dry mixing strongly influences the binder due to fibrillization, which is highly sensitive to these parameters. To improve manufacturing efficiency and reduce production costs, the mixing process should be optimized to shorten its duration. Therefore, it is necessary to develop a mixing strategy that enhances the dispersion of the conductive agent while promoting binder fibrillization, enabling the formation of an active-material microenvironment that supports both ionic and electronic transport. The fibrillization behavior of the PTFE binder is strongly influenced by the physical and chemical interactions among electrode components, including active materials, conductive additives, and solid-state electrolytes. The PTFE binder undergoes shear forces generated under external stress through adjacent particles, and the magnitude and nature of these forces directly determine the extent of PTFE fibrillization. Compared to oxide-based ceramic materials, sulfide-based solid electrolytes exhibit higher mechanical ductility, allowing them to absorb or dissipate external shear stress. This characteristic can impede the effective transmission of shear forces to the PTFE binder, potentially leading to insufficient fibrillization and, consequently, degraded mechanical integrity of the electrode film. Therefore, precise control over the properties of electrode components including particle shape, size, distribution, and surface chemistry is essential for regulating PTFE fibrillization. Furthermore, as the underlying mechanisms of this process remain largely unexplored, a systematic investigation into the complex interfacial interactions

among the electrode constituents is warranted to better understand and optimize PTFE fibrillization behavior. In parallel, improvements in equipment design are required to enhance manufacturing efficiency, promoting uniform microstructures and overcoming electron conduction limitations in the electrode.

The coating process is another critical factor for achieving a uniform electrode thickness. In conventional electrode fabrication, the density and thickness of the electrode are typically controlled by adjusting the gap during slurry coating onto the current collector. However, dry electrodes generally exhibit highly elastic properties, requiring precise control of the electrode coating process to secure reliable electrode dimensions. Furthermore, the coating process is influenced by the particle size of the active material, morphology of the conductive additive, and physical properties of the binder. The coating process, including the pressing process, determines the pore size and electrode density. Furthermore, optimization of the lamination process is required depending on the type of current collector used in the ASSB design. In coating and calendaring processes for batteries with liquid electrolytes, the applied pressure directly influences porosity and pore tortuosity within the electrode. In ASSBs, porosity should be minimized, and the microstructure should be tailored to support both ionic and electronic transport at the active material interface. In regards of the ductility of sulfide-based solid electrolytes, structural rearrangement takes place under applied pressure and during the coating step. The final electrode structure is also influenced by the composition of the electrode, particularly the presence of ionically insulating components such as the conductive agent and binder. Since conductive agents typically lack ionic conductivity and have limited deformability, strategic control over the electrode architecture is required to ensure the effective transport of both ions and electrons at the active material surface.

To achieve high-energy-density electrode designs, thicker electrodes are required for minimizing the thickness or quantity of the current collector and solid electrolyte layers. Given that dry electrode manufacturing generally involves gradual compression and thinning of the electrode mixture during fabrication, the thick electrode approach is expected to reduce the overall process burden by eliminating the need for additional thinning. While conventional high-density LIB electrodes face limitations in achieving thick coatings and maintaining performance, ASSBs

offer the possibility of improving the energy density through thick electrodes by controlling the amount of solid electrolyte to ensure efficient ion transport within the electrode. This approach, combined with advances in dry-electrode technology, will contribute to the realization of more competitive and advanced ASSB technologies.

Another critical consideration is the scalability and industrial feasibility of the dry-coating process. Recent studies have demonstrated that the dry coating process enables the fabrication of electrodes and solid electrolyte membranes with a homogeneous microstructure and superior electrochemical performance; however, its reproducibility and yield in large-scale production environments remain to be fully validated. In particular, a direct comparison with conventional wet coating processes using PVdF binders is essential for evaluating the industrial viability of the PTFE-based dry coating process. Additionally, during scale-up, Li^+ diffusion at the interface of the cathode and solid electrolyte layers should be examined carefully. In current laboratory-scale processes, high-pressure compaction is used to maximize the interfacial contact between the electrode and solid electrolyte layer. However, in large-scale manufacturing, it is crucial to determine whether a uniform pressure can be maintained and whether a consistent interfacial contact can be achieved across the entire electrode surface. Furthermore, the industry is increasingly aiming to adopt all-solid-state systems that do not require external pressure; therefore, it is essential to determine the ways the dry coating process can be adapted to such pressure-free ASSB systems.

In addition to the aforementioned interfacial challenges, the issue of particle segregation during the dry mixing process should also be carefully considered. As the electrode composition consists of multiple solid components with various particle sizes and densities, there is a high risk of particle segregation during the dry mixing process, which results in an inhomogeneous distribution of the active material, conductive additive, and binder within the dry mixture. While laboratory-scale research has primarily focused on addressing these issues through material-level strategies, such as incorporating multiple types of conductive additives or engineering surface characteristics of particles to manipulate interparticle interaction, these approaches face inherent limitations in scalability and practical implementation at the industrial scale. Therefore, further studies on exploring dry mixing processing and equipment parameters are essential. In particular, understanding the influence of dry mixing conditions—such as rotational speed, energy input, shear force, mixer geometry, and mixing sequence—on compositional uniformity is crucial. Given the complexity and scale of such investigations, interdisciplinary collaboration between materials science, mechanical engineering, and industrial partners is indispensable for advancing scalable dry mixing technologies and developing high-throughput systems capable of producing dry mixed powder with uniform distribution of electrode components.

Moreover, the fabrication of free-standing films via the roll-to-roll dry coating process introduces additional challenges related to yield and mechanical reliability. During continuous production, the edge regions of the electrode films are typically removed to guarantee dimensional uniformity, resulting in unavoidable production loss. To address this issue, it is necessary to study for reducing-edge waste and recovering trimmed-off segments

of the electrode film for reuse, as well as to improve roll-to-roll equipment designs that minimize edge-related production loss. To obtain reliable and defect-free free-standing films during high-speed processing, the free-standing films should be mechanically robust. However, the criteria for the specific mechanical properties required for roll-to-roll processing have not yet been clearly understood, and variations in equipment configurations across production lines present additional challenges for standardization. Consequently, an interdisciplinary collaboration involving materials science, mechanical and process engineering, systems engineering, and close partnerships with industry is also essential to achieve clear mechanical performance benchmarks and to co-develop roll-to-roll processing technologies that ensure both high productivity and reproducibility in large-scale ASSB manufacturing.

In conclusion, although the dry coating process has significant technological and economic advantages over the conventional wet coating process for ASSB fabrication, further validation of its scalability and industrial feasibility is essential. Future advancements in material engineering and process optimization are also crucial for unlocking the potential of the dry coating process for high-energy-density ASSBs and accelerating their commercialization.

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Conflict of Interest

The authors declare no conflict of interest.

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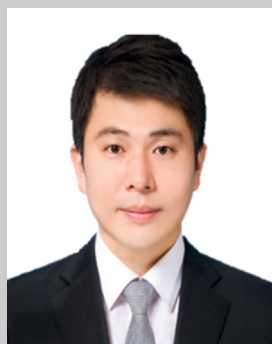
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